

Computational Method in Chemical Engineering T2730C2

Textbooks:

- Lorens T. Biegler, E. Ignacio Grossmann, Arthur W. Westerberg, Systematic Methods of Chemical Process Design, Prentice Hall International.
- Warren D. Seider, J. D. Seader, Daniel R. Lewin, Product and Process Design Principles: Synthesis, Analysis, and Evaluation, 2nd Edition, Wiley.
 - Robin Smith, Chemical Process: Design and Integration, Wiley.
- James M. Douglas, Conceptual Design of Chemical Processes, McGraw Hill International, 1988.

Unit 5 – Energy Integrated Distillation Process

- Heat Integrated Distillation Trains
- Impact of Pressure
- Multi Effect Distillation
- Heat Pumping, Vapour Recompression and Reboiler Flashing
- Positioning of Heat Engines and Heat Pumps

Heat Integrated Distillation Trains: Impact of Pressure

- **The column pressure of a distillation column is a key design variable because it determines the temperature levels in the reboiler and condenser, and consequently, the possible heating and cooling media.**
- The principal vehicle for enabling the kinds of energy exchanges in most processes is through the adjustment of the pressure of the distillation towers, although in many cases it is possible to operate the reactor at a higher temperature.
- **The other alternative is to reduce the pressure of the distillation tower, thereby reducing the reboiler temperature and permitting the condensation of the product in the reboiler of the first tower.**
- This reduces the usage of cooling water and steam, or eliminates them entirely, depending upon the cooling and heating demands.
- **Also, as the distillation pressure is reduced, the separation is made easier and the number of stages is decreased.**

Heat Integrated Distillation Trains: Impact of Pressure

- On the down side, however, the temperature of the condenser is reduced.
- **If it is reduced below the temperature at which cooling water can be used, the cost of refrigeration becomes a significant cost factor (largely through an increased compression load).**
- Furthermore, the integrated process is more difficult to control.
- In many cases, however, the combined savings in the utilities and the capital cost of the heat exchangers and the column exceeds the added refrigeration costs, and a reliable control system can be designed for the integrated process.

Heat Integrated Distillation Trains: Impact of Pressure – T-Q diagram

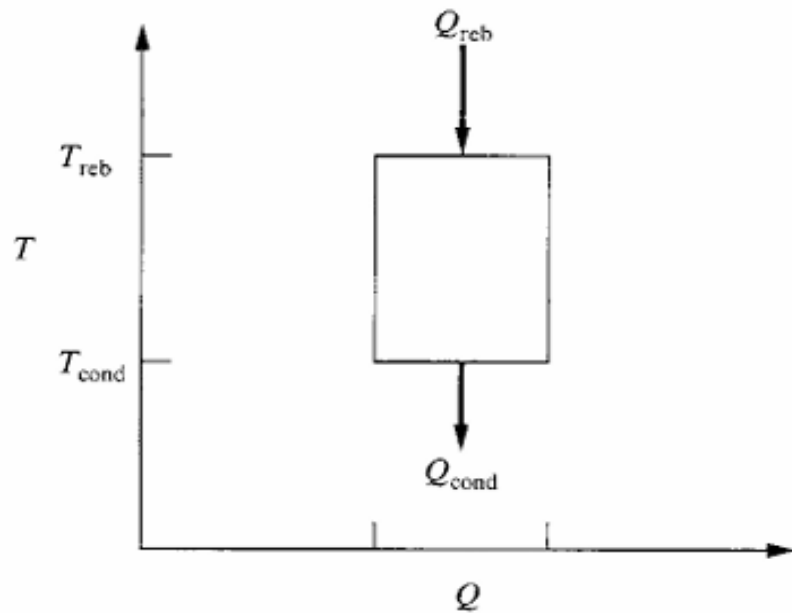
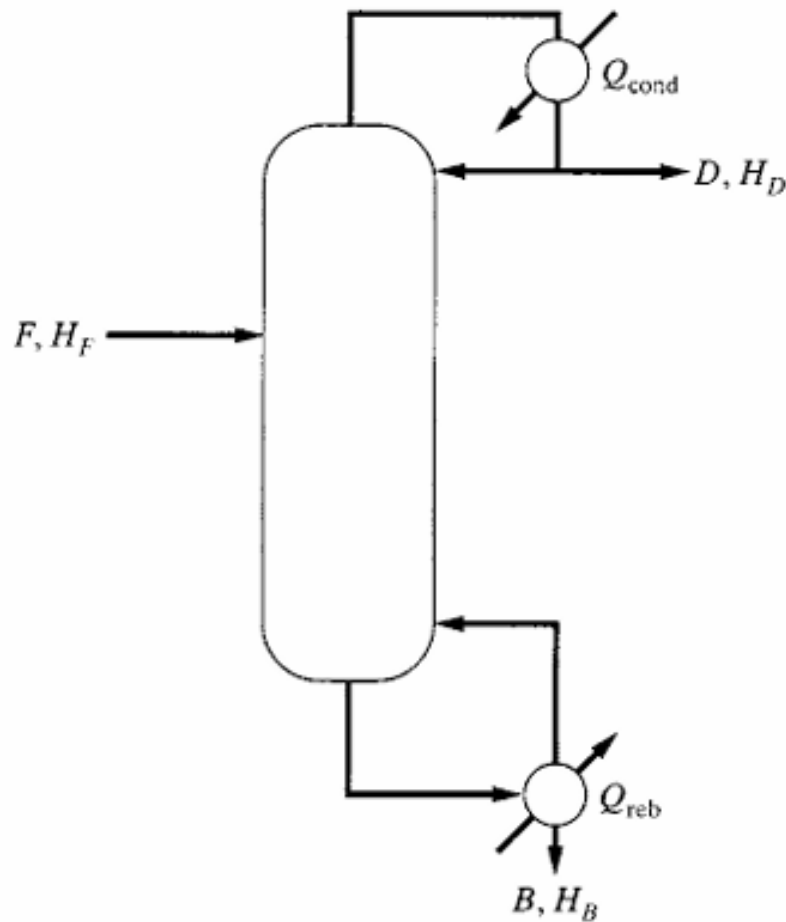


Figure 10.40 T-Q diagram for a distillation column.

- When working with the composite heating and cooling curves for a process, it helps to examine the heating and cooling requirements for a distillation column **using a T-Q diagram**.
- This diagram shows the heat provided to the reboiler, Q_{reb} at temperature, T_{reb} .
- Similarly, the diagram shows the heat removed from the condenser, Q_{cond} at temperature, T_{cond} .
- The principal **assumption** in Figure is that $Q_{reb} = Q_{cond}$.
- Reasonable for most distillation towers, especially when the feed and product streams are saturated liquids.

Heat Integrated Distillation Trains: Impact of Pressure



- Energy balance in the steady state is

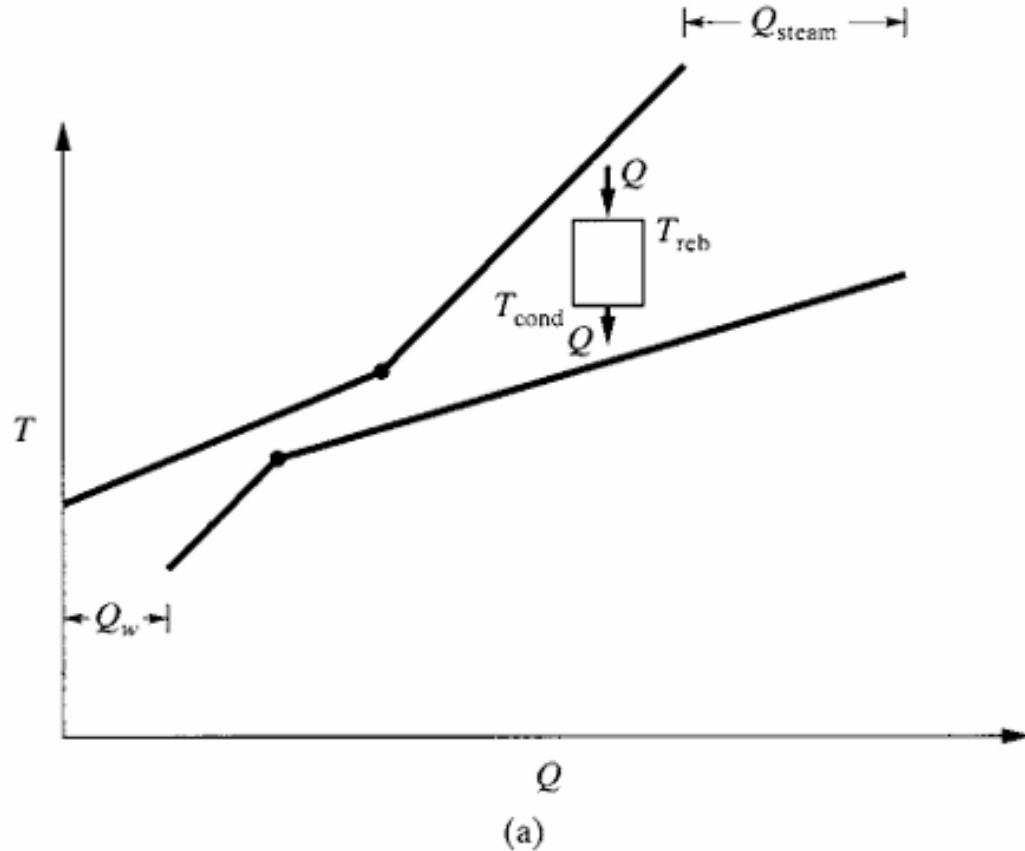
$$FH_F - DH_D - BH_B + Q_{\text{reb}} - Q_{\text{cond}} = 0$$

- Then, for saturated liquids, where $T_{\text{cond}} < T_F < T_{\text{reb}}$ a reasonable approximation is:

- $$FH_F - DH_D - BH_B \approx 0$$

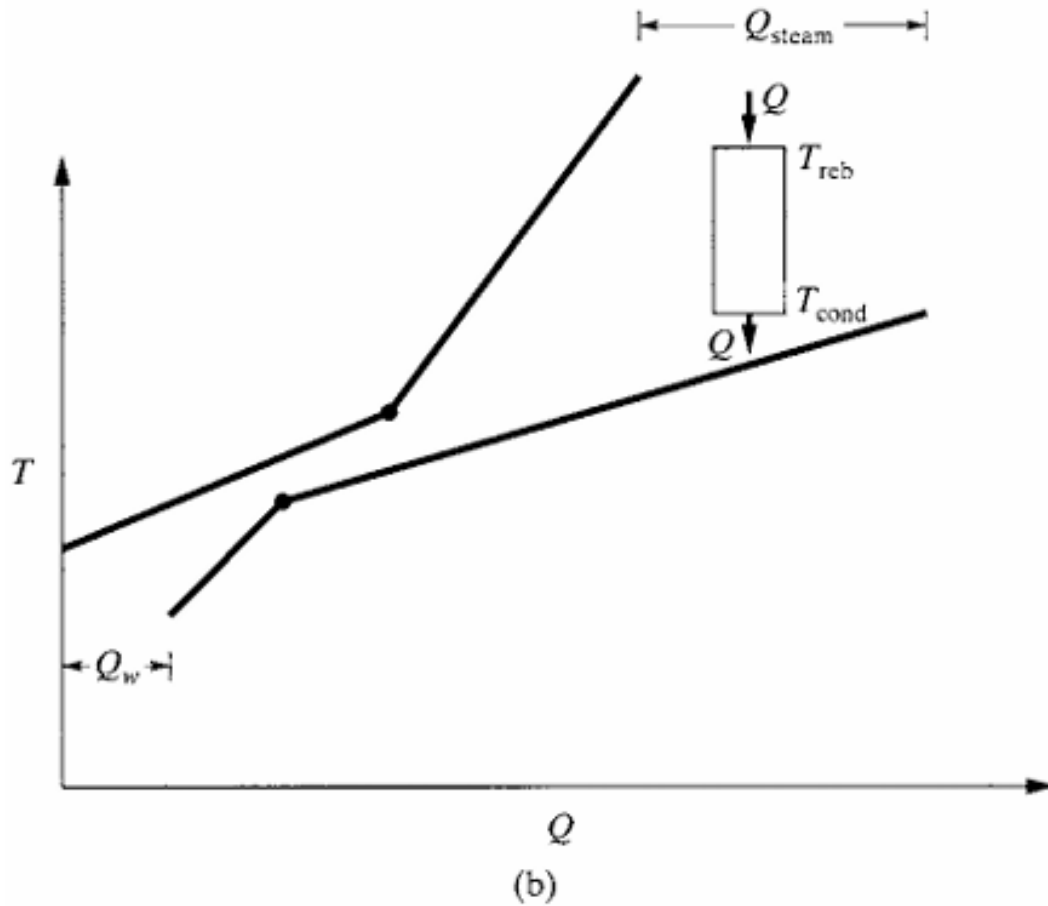
$$Q_{\text{reb}} \approx Q_{\text{cond}}$$

- Heat duty, $Q = Q_{\text{reb}} \approx Q_{\text{cond}}$ is related directly to the reflux ratio.
- When **Q is reduced, because the cost of fuel is high, the number of trays (or the height of the packing) increases.**
- Trade-offs between operating and capital costs significantly influence the optimal design.



Positioning distillation towers between hot and cold composite curves: (a) exchange between hot and cold streams

- The composite hot and cold curves are created without including the heat duties of the reboiler and condenser for a potential distillation tower.
- **When possible, the pressure level of the tower is adjusted to position its T - Q rectangle to lie below the hot composite and above the cold composite curves.**
- In this way, heat can be transmitted from the hot process streams to the reboiler and from the condenser to the cold process streams without increasing the external utilities.
- **It is possible for the reboiler to accept energy from hot streams above the pinch and for the condenser to reject heat to the cold streams below the pinch.**
- In this case, **heat flows across the pinch**, with the consumption of the hot and cold utilities increasing by Q .
- Clearly, when the cost of fuel is high, this design is not recommended.



Positioning distillation towers between hot and cold composite curves: (b) exchange with cold streams.

- Unfortunately, it can be difficult to position a tower as shown in (a) unless the chemical species being separated are close-boiling (e.g., the split between propylene and propane).
- **Alternatively, the pressure of the tower can be adjusted and the tower positioned so that the reboiler receives its energy from a hot utility and the condenser rejects its energy to the cold process streams above the pinch, as shown in (b).**
- This **reduces the load of the hot utilities** required by the HEN for the remainder of the plant, while satisfying the heating demand of the reboiler.
- When the cost of fuel is high, this provides an attractive design.

Multi Effect Distillation

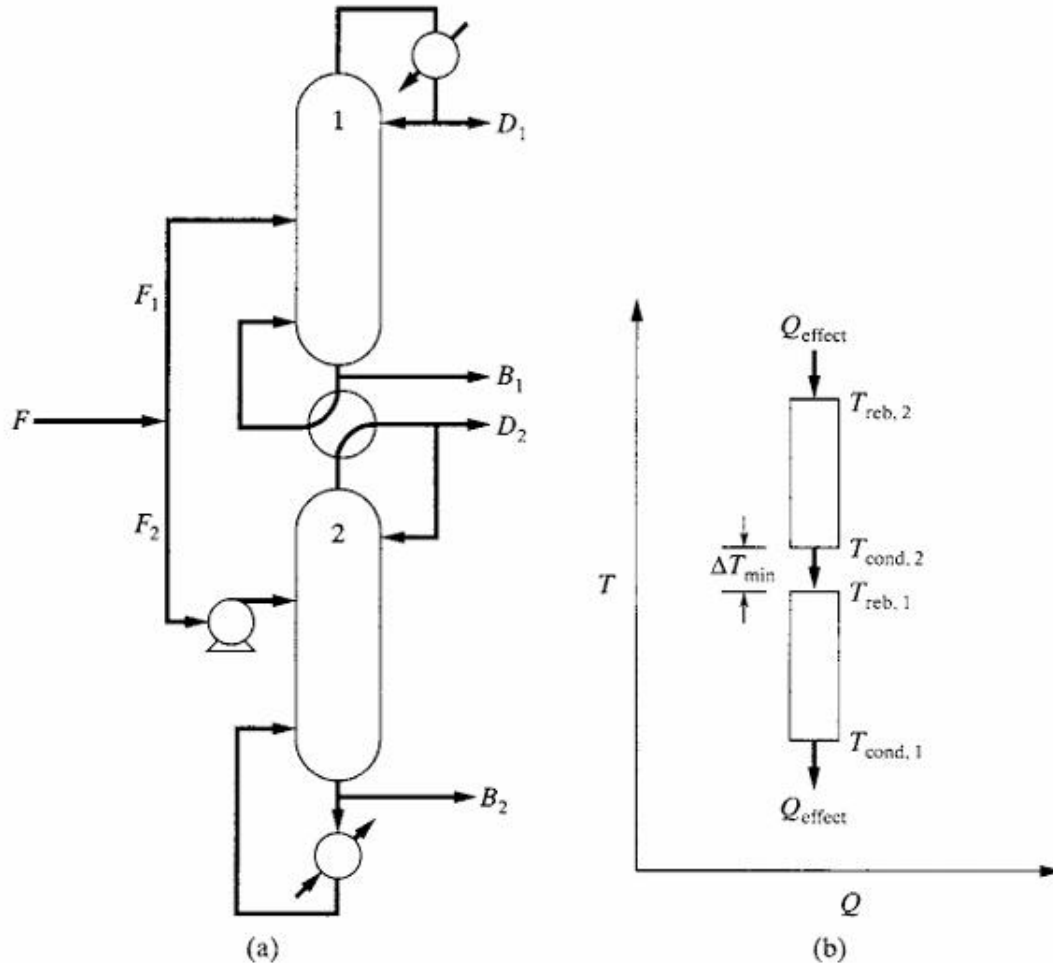


Figure 10.43 Two-effect distillation: (a) tower and heat exchanger configuration; (b) T - Q diagram.

- For separations where the T - Q rectangle for a distillation tower cannot be positioned under the hot composite and above the cold composite curves, several possibilities exist for creating a more energy-efficient distillation operation when the price of fuel is high.
- One widely used configuration for distilling water is **multiple-effect distillation**, in which the feed stream is split into approximately equal portions and sent to the **same number of separate distillation towers, each operating at a different pressure**.
- The pressures in the towers decrease from the bottom to the top of the cascade so that the temperatures of the adjacent condensing vapors and boiling liquids differ by ΔT_{min} .

Multi Effect Distillation

- In this way, heat from the condensing vapor in the tower below, at a pressure of P2 is transferred to boil the bottoms liquid from the tower above, at the lower pressure of P1.
- The flow rates of the feed streams are adjusted to equate the duties of the adjacent condensing and boiling streams.
- In this way, the heat duties associated with condensation and boiling $Q_{\text{effect}} \approx Q/N_{\text{effect}}$.
- When *the price of fuel is high, this represents a substantial reduction in the operating costs.*
- On the down side, however, the pressure level is increased in $N_{\text{effect}} - 1$ of the distillation towers, and **multiple towers are needed.**

Multi Effect Distillation

- While **pumping costs to increase the pressure of liquid feed streams are low**, the tower walls are thicker, **increasing the purchase costs**, and the relative volatility is decreased at higher pressures, **increasing the number of trays** (or height of the packing) required to maintain the same reflux (Q_{effect}).
- In many cases, ***these increases in the operating and purchase costs are small when compared with the large savings in the utilities for condensing and especially boiling.***
- The net effect of dividing the feed stream into N_{effect} nearly equal portions is to **elongate the T - Q diagram** as shown in (b).
- The total area is not conserved because the pressure level in each effect determines the bubble-point temperatures of its condensing vapor and boiling liquid streams.

Multi Effect Distillation

(a): feed-splitting (FS)

(b): light split/forward heat-integration (LSF) configuration

- The feed is pumped and sent to the high-pressure column, entire feed stream is sent to the first tower.
- About half of the light key component is removed in the distillate at high purity.
- The bottoms product, which contains the remainder of the light key component, is fed to the low-pressure column.
- Heat integration is in the direction of mass flow.

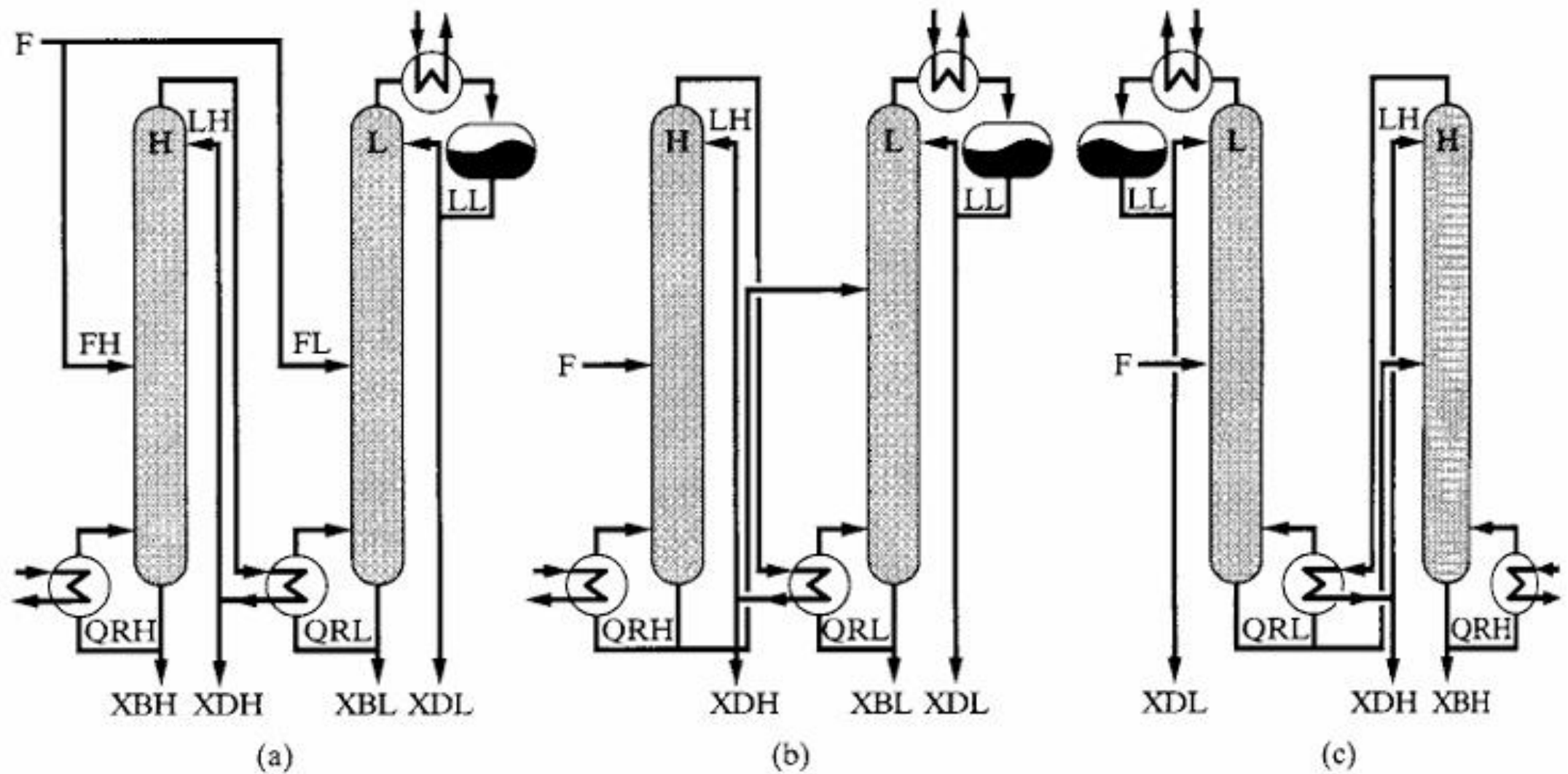


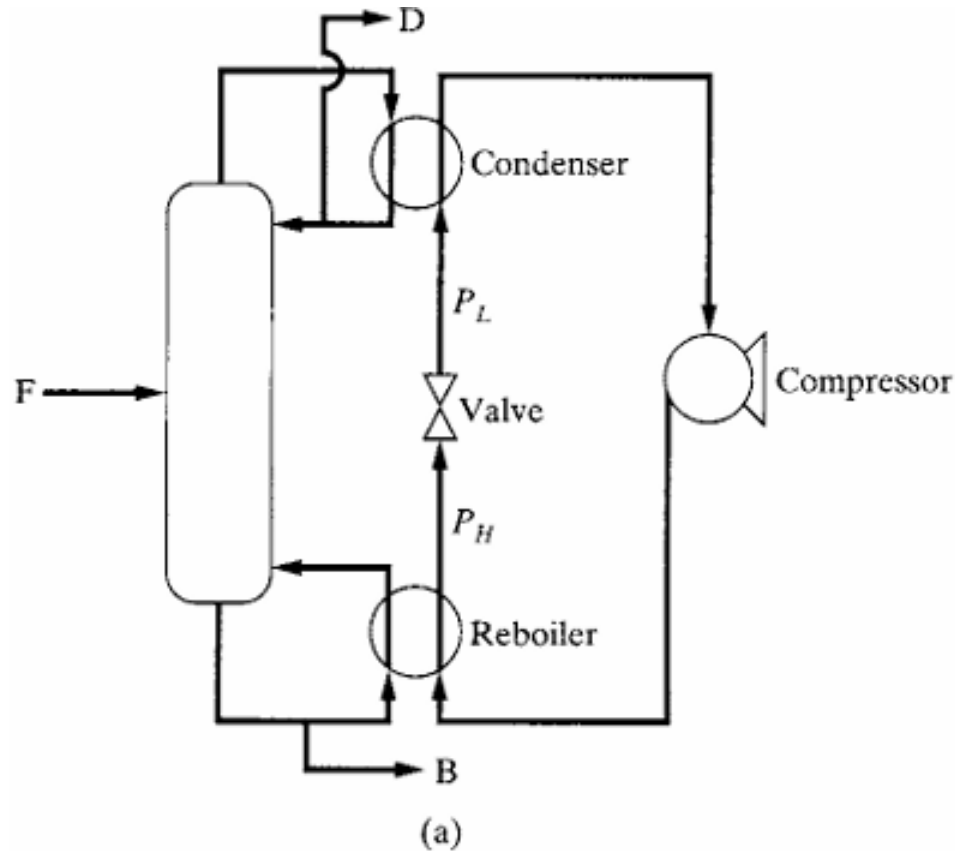
Figure 10.44 Variations on two-effect distillation: (a) FS; (b) LSF; (c) LSR.

Multi Effect Distillation

(c): light split/reverse heat integration (LSR) configuration

- The feed is sent to the low-pressure column, entire feed stream is sent to the first tower.
- Half of the light key component is removed in the distillate, with the bottoms product pumped and sent to the high-pressure column.
- Heat integration is in the reverse direction of the mass flow.
- Configurations are compared when operation is in the steady state.
- Then, the controllability and resiliency (C&R) of each configuration is assessed in response to typical disturbances, and verified by dynamic simulations of the FS and LSR configurations, using HYSYS.Plant, confirming the findings of the C&R analysis.

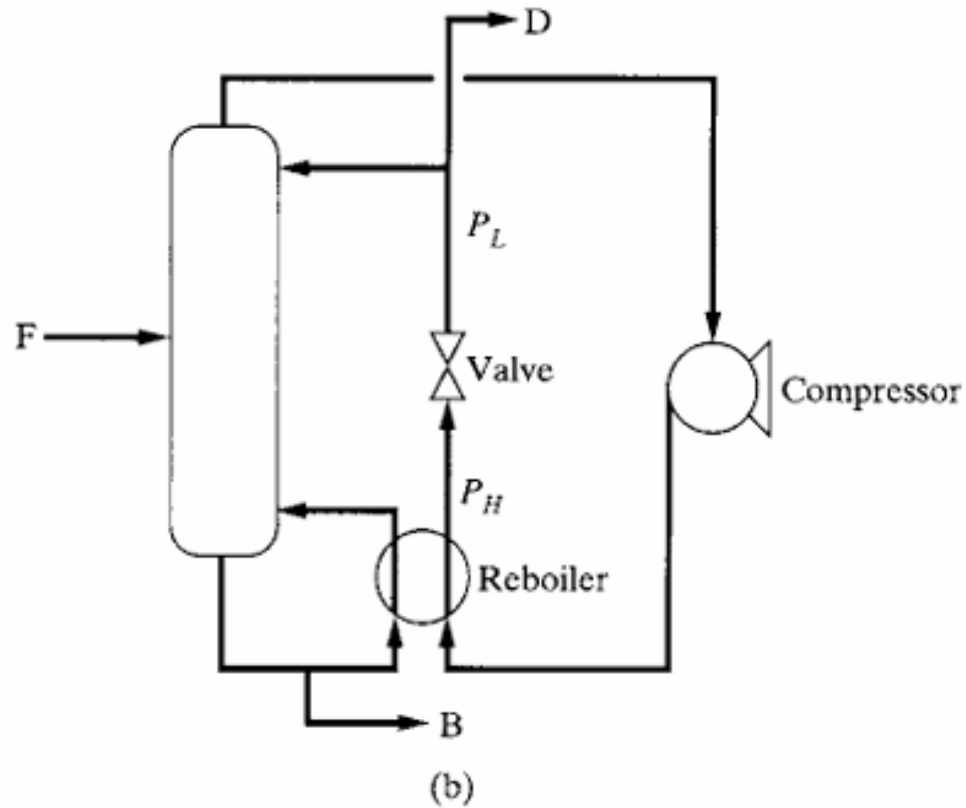
Heat Pumping, Vapour Recompression and Reboiler Flashing



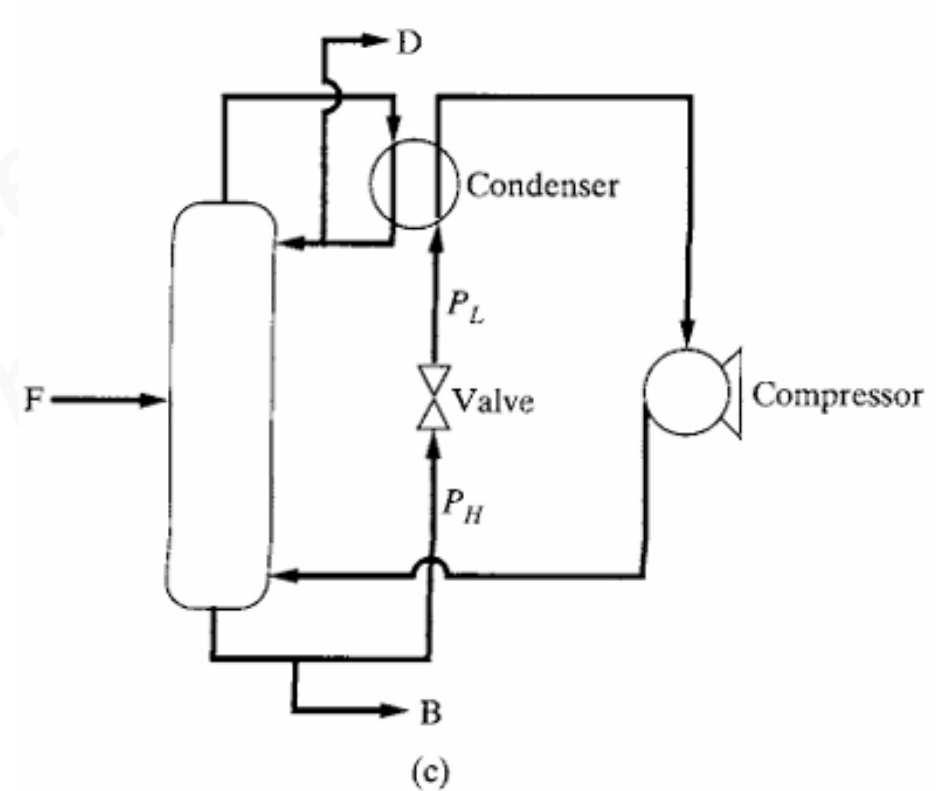
(a) Heat pumping

- Other more sophisticated configurations, designed to increase the thermodynamic efficiency when the price of fuel is high, **permit the vapor overhead to be condensed with the bottoms liquid from the same distillation column.**
- ***Each of three configurations, heat-pumping, vapor recompression, and reboiler flashing , involves expensive compression of a vapor stream.***
- The heat pump operates like a refrigeration cycle and requires an external fluid as the working medium.
- It pumps available heat from a low-temperature level up to a higher temperature level where it can be used more effectively.

Heat Pumping, Vapour Recompression and Reboiler Flashing



(b) Vapor compression



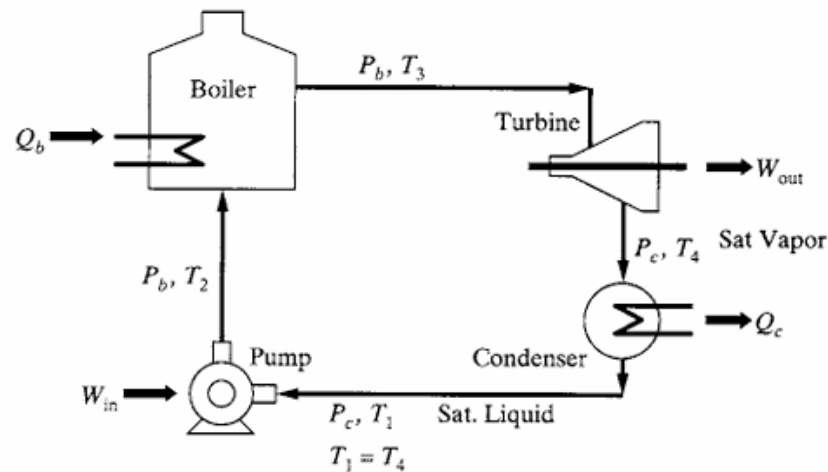
(c) Reboiler flashing

Heat Pumping, Vapour Recompression and Reboiler Flashing

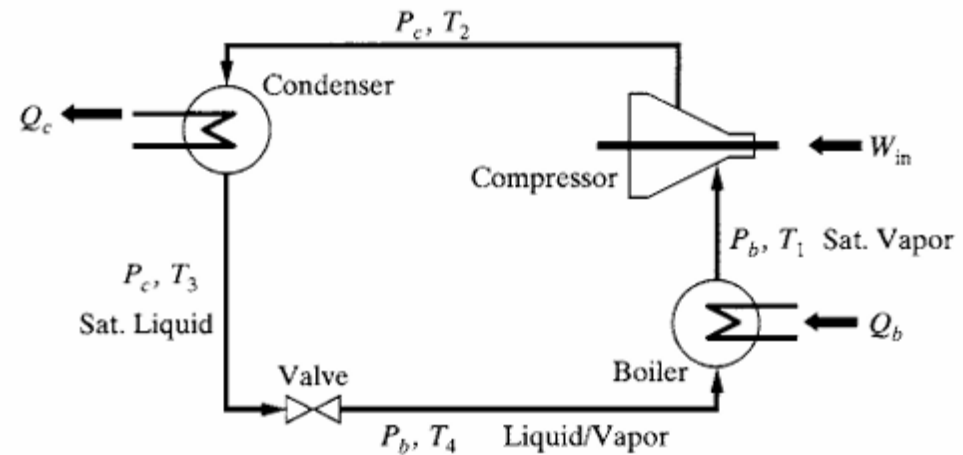
- The other two configurations do not have external working fluids. Instead, they use the internal process fluids.
- **To be effective, the savings in the cost of utilities and purchase costs for the heat exchangers must be greater than the increased utility and capital costs associated with the compressor.**
- Note that these configurations are most attractive for the **separation of close-boiling mixtures** because relatively small pressure changes are required, and consequently, the **cost of compression is not too high.**

Positioning of Heat Engines and Heat Pumps

- When processes have significant power demands, usually in compressor loads, it is normally sound practice to **operate at or near the minimum utilities for heating and cooling**.
- This is because the **annualized cost is dominated often by the operating and capital costs associated with satisfying these power demands**.
- Given the desirability of operating these processes at minimum utilities, Townsend and Linnhoff (1983a, b) make recommendations concerning the positioning of heat engines and heat pumps.



Heat Engine



Heat Pump

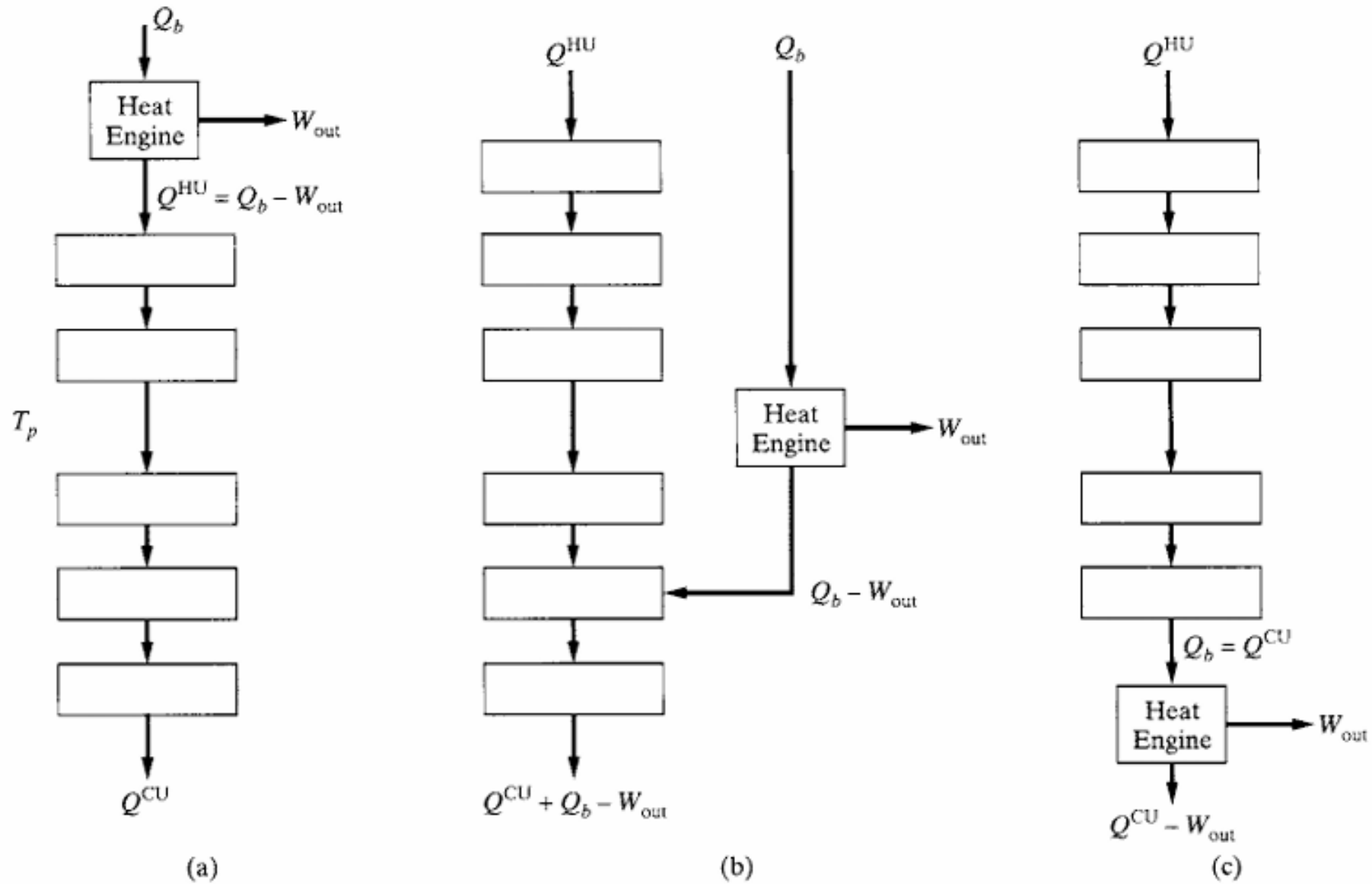


Figure 10.48 Alternatives for the placement of heat engines: (a) above T_p ; (b) across T_p ; (c) below T_p .

Positioning of Heat Engines

- (a): heat engine is positioned above the pinch, to satisfy the demand for hot utilities Q^{HU} . Q_b is required by the boiler and the net power produced is W_{out} , neglecting the small power requirement of the pump. Hence, W_{out} is produced by adding the equivalent heat to Q^{HU} .
- (b): heat engine accepting heat above the pinch and rejecting heat below the pinch. The total utilities above the pinch are incremented by Q_b , and below the pinch, the cold utilities are incremented by $Q_{b-W_{out}}$. When the cost of fuel is high, this configuration is less profitable than configurations with heat engine entirely above or below pinch.
- (c): heat engine is positioned below the pinch, the heat that would be rejected to cold utilities, Q^{CU} is sent to the boiler. W_{out} is recovered from the turbine and the remainder is rejected to the cold utilities.

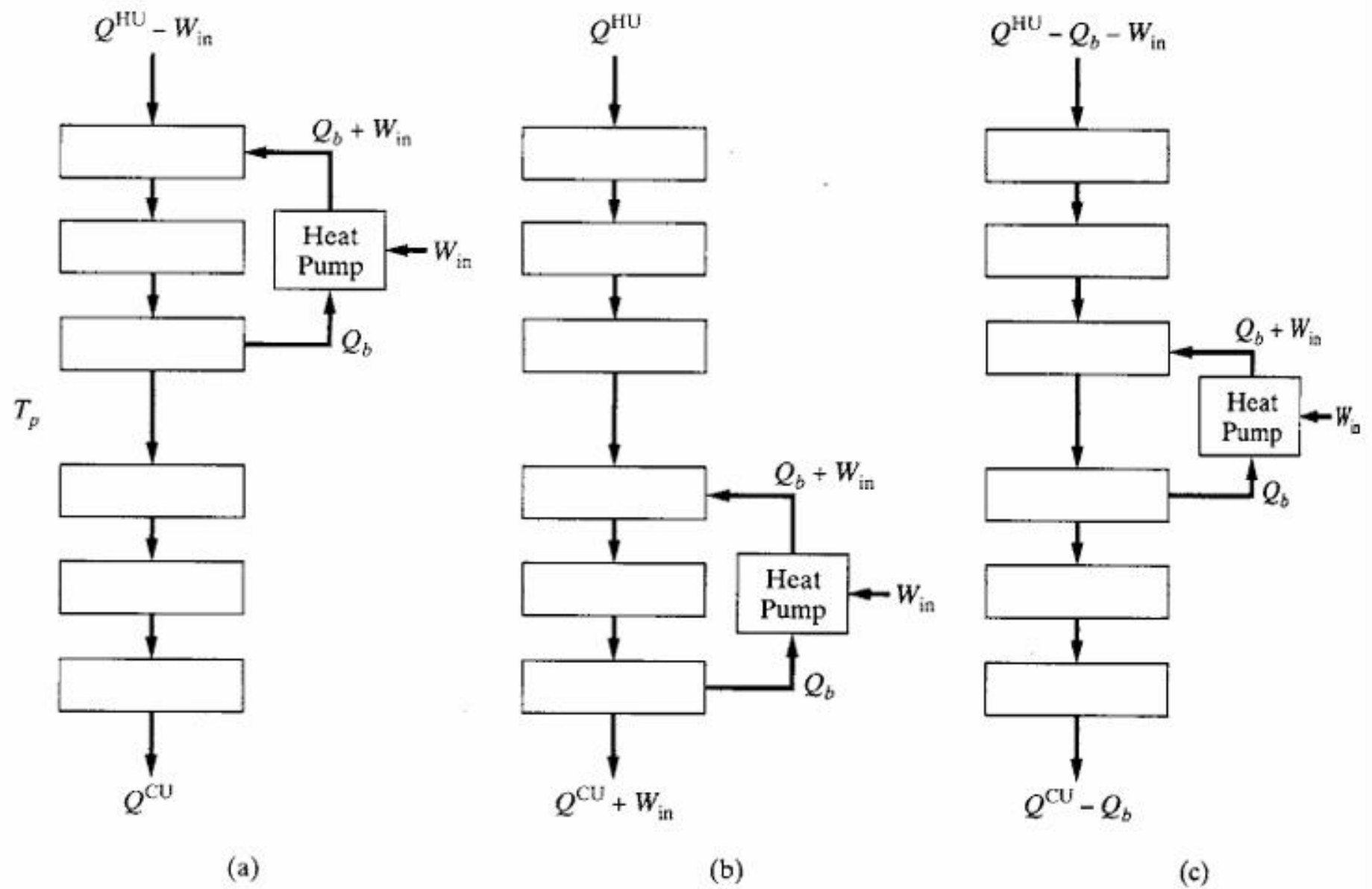


Figure 10.50 Alternatives for the placement of heat pumps: (a) above T_p ; (b) below T_p ; (c) across T_p .

Positioning of Heat Pumps

- **(a): heat pump is positioned above the pinch**, its compression load, W_{in} reduces the hot utility load by this amount but does not reduce the cold utility load below the pinch. In this case, expensive power is converted directly to less valuable heat to reduce the hot utility load.
- **(b): heat pump is positioned below the pinch**, its compression load, W_{in} increases the cold utility load by the same amount without affecting the hot utility load. This is less desirable than when the heat pump is positioned across the pinch, where both the hot and cold utilities are decreased.
- **(c) heat pump is positioned across the pinch**, heat is removed from a temperature interval below the pinch and rejected to a temperature interval above the pinch, causing a reduction in both the hot and cold utility loads but at the expense of shaft work.

Positioning of Heat Engines and Heat Pumps

Townsend and Linnhoff Heuristics :

- When positioning heat engines, to reduce the total utilities, place them entirely above or below the pinch.
- When positioning heat pumps, to reduce the total utilities, place them across the pinch.

Following the heuristics of Townsend and Linnhoff, **optimal design strategies** have been developed (Colmenares and Seider, 1989a, b). These involve the following steps:

1. **Carry out the temperature-interval method** to locate the pinch temperatures and the minimum hot and cold utility loads.

Positioning of Heat Engines and Heat Pumps

2. **Lump the temperature intervals together** above the pinch to create intervals having **heat deficits**. Intervals in which more heat is required to heat the cold streams than is available from the hot streams to be cooled.

Similarly, lump the temperature intervals together below the pinch to create intervals having negative heat deficits (intervals in which more heat must be removed from the hot streams than can be consumed by the cold streams.)

3. Create a **superstructure that includes the candidate heat engines and heat pumps**. These can add heat to the lumped intervals above the pinch and remove heat from the lumped intervals below the pinch.
4. **Formulate a NLP** to minimize the total annualized cost of the heat engines and heat pumps. The design variables include the pressure levels and the flow rates of the working fluids in the heat engines and the heat pumps.

Positioning of Heat Engines and Heat Pumps

The constraints include the heat balances for the condensers and boilers, the heat balances for the temperature intervals, an energy balance to satisfy the power demand, bounds on the heat removed or added to the temperature intervals, and bounds on the temperatures and pressures.

5. Solve the NLP using a solver such as MINOS within GAMS.

Quick Recap

- Why is the column pressure of a distillation column is a key design variable?
- What is a T-Q diagram?
- Why assume $Q_{\text{reb}} \approx Q_{\text{cond}}$?
- What are the possible positions of distillation towers between hot and cold composite curves?
- What is multi effect distillation? What are its types?
- Why use heat pumping, vapor recompression, and reboiler flashing?
- What are the possible positions of Heat Engine and Heat Pumps in HEN?
- What are the Townsend and Linnhoff's Heuristics for positioning of heat engines and heat pumps?
- What are some steps of the optimal design strategy?